fit@hcmus

Software Testing

Automation Testing

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Topics covered

- **Overview of test automation**
- Typical test automation process
- Testing approaches
- Automation tools

Test automation

- Test automation refers to the use of software tools to execute tests
- Automated testing tools can enter data, run tests, compare results, and report test results
- Automated testing vs. manual testing
  - Manual testing: tests are performed by humans
  - Automated testing: tests are performed by computers

Why Test Automation?

- Manual testing is time and cost consuming
- Automation testing shorten tests and project duration
- Difficult to do manual testing in some situations
  - Multi-lingual sites
  - Performance test
  - Security test
- Automation helps increase test coverage
- Manual testing can become tedious and error prone

Benefits of Test Automation - 1

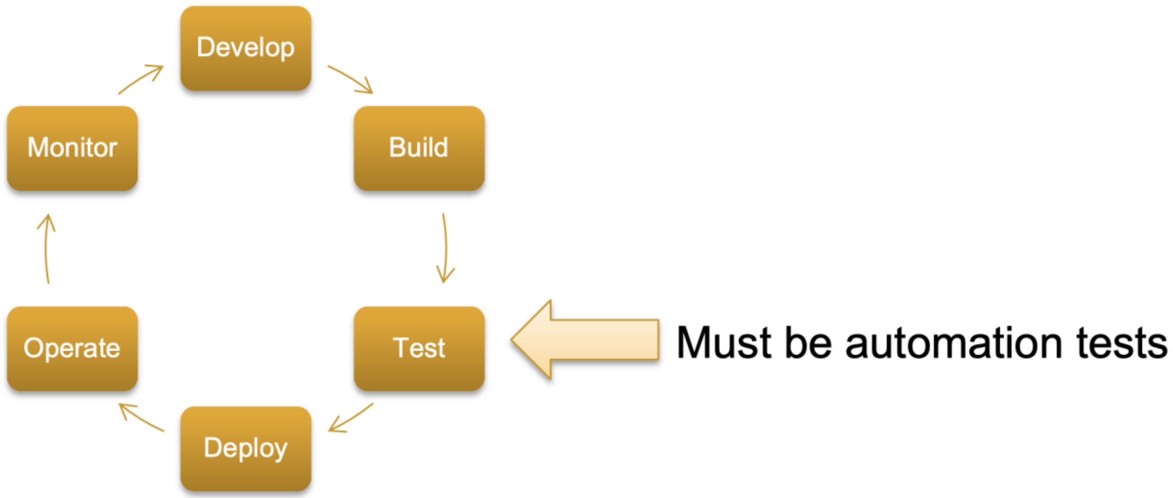
- Saves Time and Cost
- Faster than the manual testing
- Early time to market
  - Better speed in executing tests
- Reusable testing

Benefits of Test Automation - 2

- Wider test coverage of application features
- Reliable in results
- Improves accuracy
- Test more frequently and thoroughly

Test automation is essential for DevOps

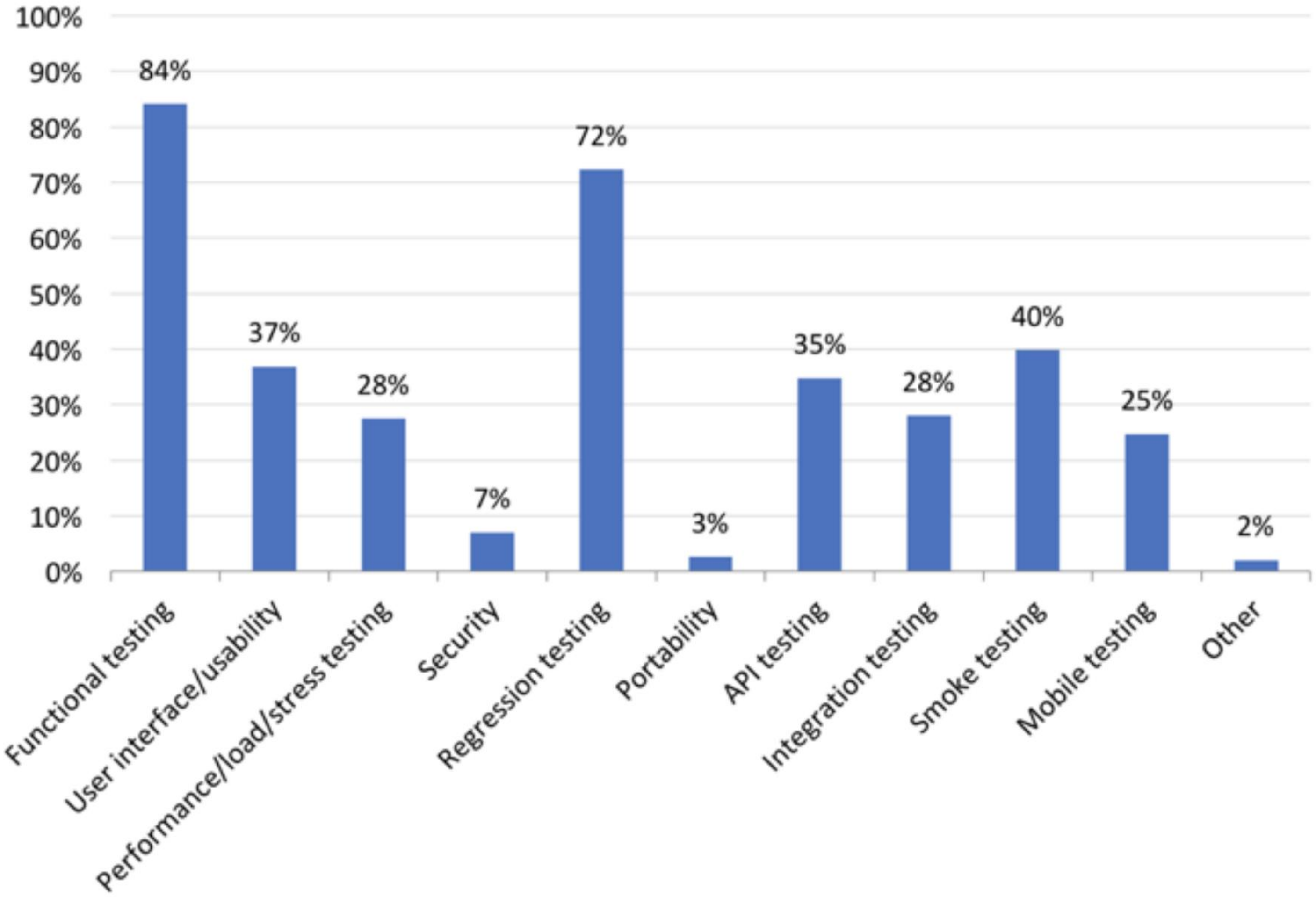
- DevOps – a practice for making steps from Development to Operation easy and quick
- DevOps reduces software lifecycle time significantly
- It becomes a trend today



When Test Automation Works Best?

- Test automation works best for some testing
  - High risk, business critical test cases
  - Test cases that are executed repeatedly
    - Such as regression test
  - Test cases that are very tedious or difficult to perform manually
  - Time consuming test cases
  - Performance tests
  - Security tests

Testing types using automation

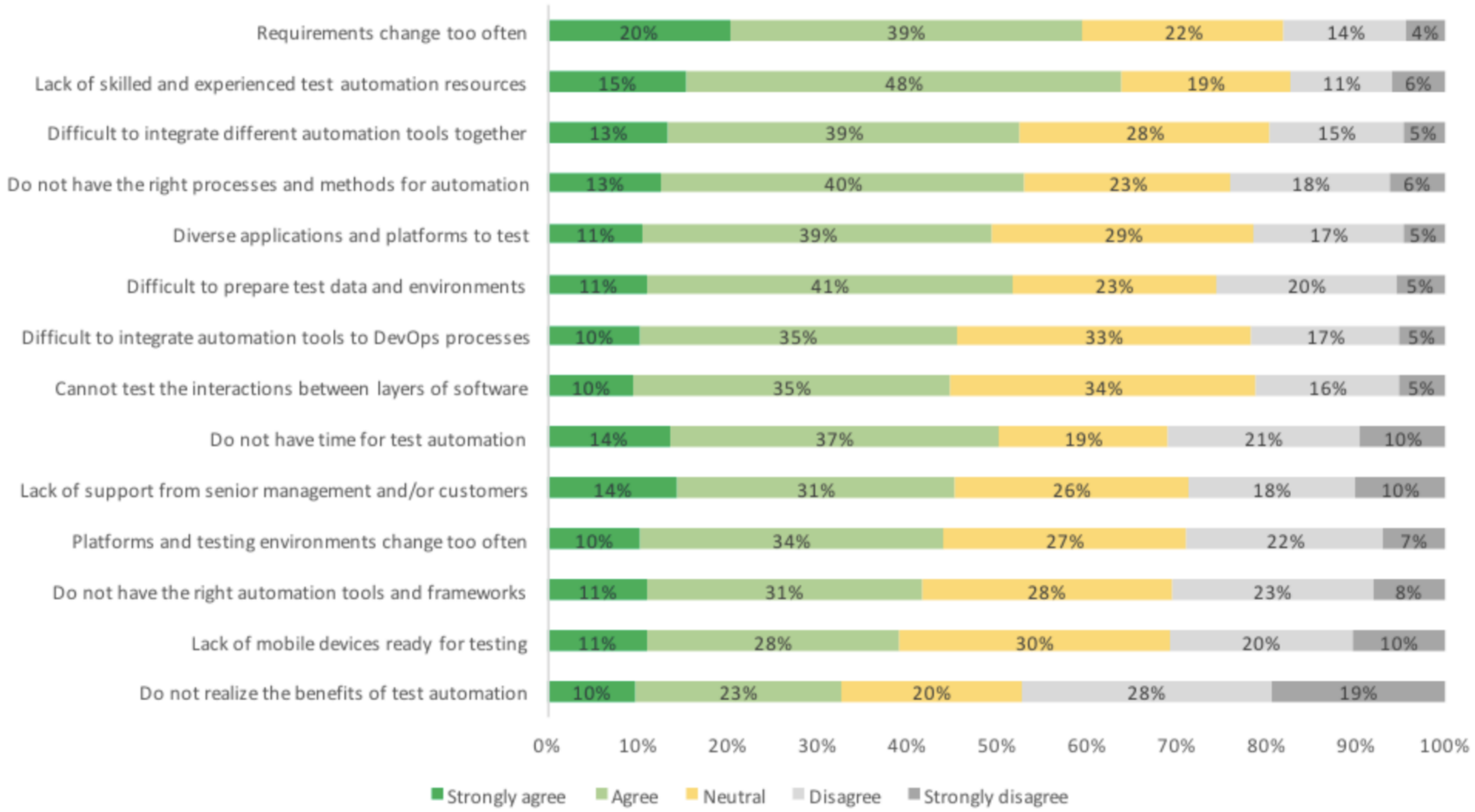


Source: “The most striking problems in test automation: A survey”, 2018. Katalon.com

When Test Automation Not Suitable?

- Newly designed test cases
  - Test cases should be manually tested at least once
- Test cases for frequently changed requirements
- Ad-hoc test cases
- UI/UX testing
  - Dependent on human judgment and experience

Challenges of test automation



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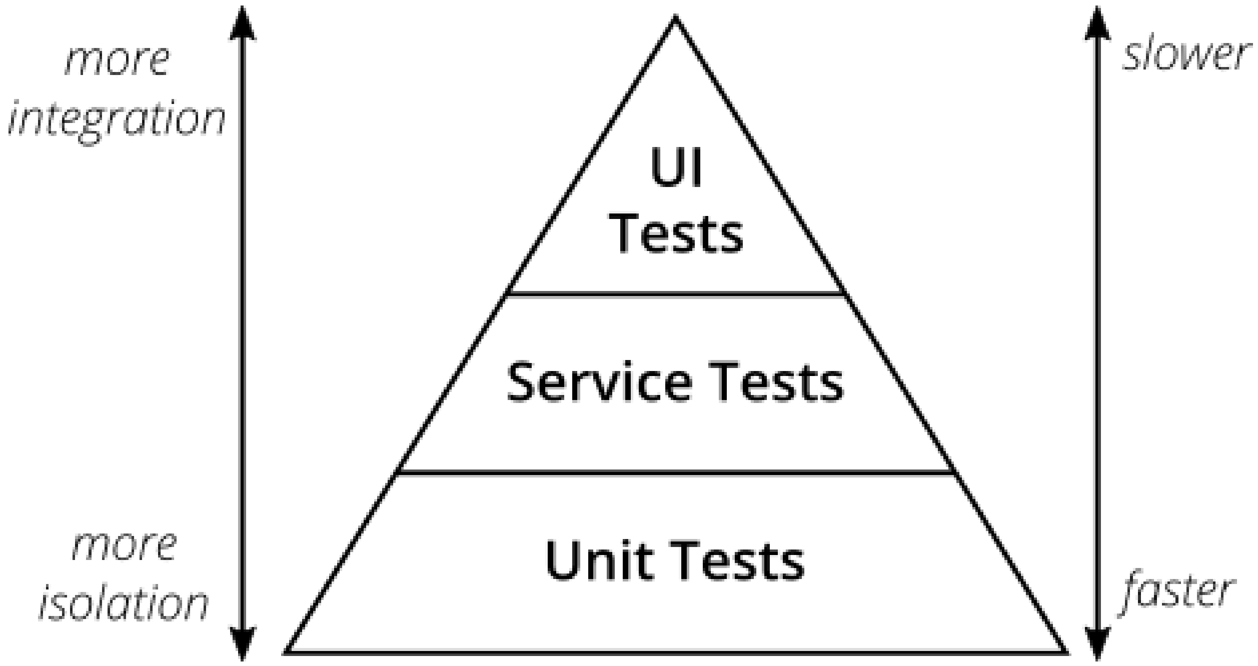
Levels of test automation

- Unit testing
  - Methods, functions, classes
- Integration testing
  - Integration multiple parts
- System testing
  - Focusing on UI, user's features



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- **Typical test automation process**
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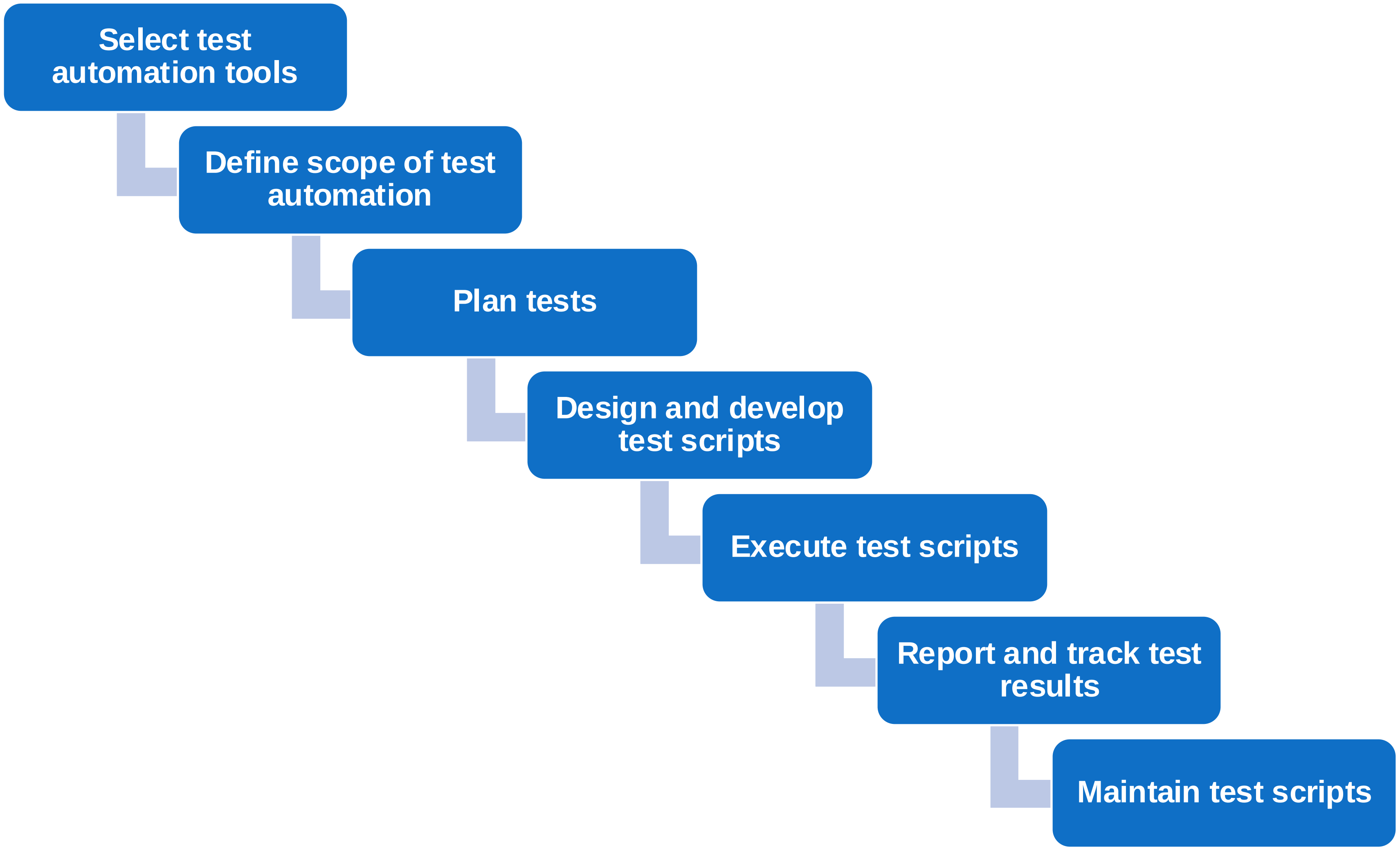
Typical Test Automation Process



Select Automation Tools

- Selecting tools suitable for applications under test (AUT) is very challenging
- Types of tool
  - Commercial: some are powerful but expensive
  - Open source: free but limited functionality
- Criteria to evaluate
  - Budget
  - Easy of use
  - Scripting languages
  - Platform (Windows, Unix, Mac OS, etc)
  - Training
  - Team experience

Define Scope of Test Automation

- Which areas in AUT to be automated
- Which areas to be tested manually
- Areas to be considered
  - Business important features
  - Scenarios which have large amount of data
  - Common functionalities across applications
  - Reused business components
  - Complexity of test cases
  - Test cases for cross browser testing

Plan tests

- Define test plans and strategies for testing
  - Tools to be used
  - Test approaches
    - Functional, non-functional, usability, performance, security testing,...
    - Automation testing approaches
    - Unit testing, system testing, integration testing, acceptance testing
  - Schedule and timeframe
  - Staffing
  - Strategies for testing
    - Manual testing vs. automation testing, areas to test, test environments, etc.

Design and Develop Test Scripts

- Design test cases
- Design test data
- Design and develop test framework and test scripts
- Evaluate test scripts
- Similar to programming
- Test framework: a set of rules for automation
- Test framework consists of software systems, function libraries, reusable modules, data source, etc.

Execute Tests, Report and Track Test Results

- Run test scripts on AUT
- Test execution is usually controlled by the automation tool
- Test results are compared with expected results
- Test results are reported
- Defects are captured and entered to defect management systems

Maintain Test Scripts

- Test scripts have to be maintained often as software changes
  - New functions are added, updated
  - Requirements are changed
- As developers change source code, test scripts may not work anymore
  - Parameters are changed
  - GUI objects are changed
  - Outputs are changed
- Maintaining test scripts can be very time consuming

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Scripting Approaches

- Record and playback
- Linear scripting
- Modular scripting
- Data-driven testing
- Keyword-driven testing

Record and Playback [1]

- Tools record users' actions performed on AUT
- Often, scripts are generated after recording
- Tools playback what is recorded
- Popular feature in many automation tools

Record and Playback [2]

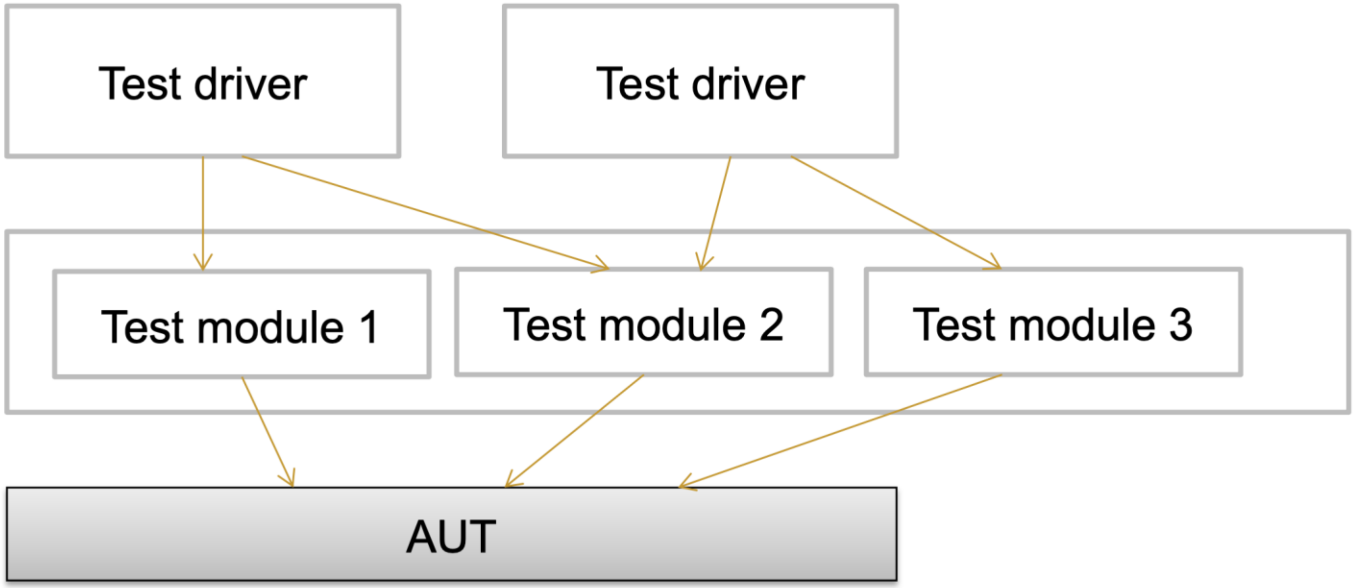
- Advantages
  - Easy to use
  - No programming skills needed
  - Good for learning
- Problems
  - Scripts can only be created when AUT is ready
  - Does not test anything. Checkpoints are needed to test
  - Only suitable with UI testing
  - Small changes in UI can cause tests to stop
  - Hard to manage and maintain
    - Lots of test scripts are created
- Not a good approach for advanced testing

Linear Scripting

- Scripts are created to test AUT
- Use some programming languages
- Scripts are also created with Record and Playback
- A test project can have many test suites
  - A test suite has one or more test cases
- Good for simple test cases
- Difficult for large automation

Modular Scripting

- Place test scripts into functions or modules
- Use drivers to call these functions or modules to execute scripts on AUT



Data-Driven Testing [1]

- Data is separated from test scripts
- Test scripts read data automatically from files
- Can test multiple data items, different inputs
- Test driver scripts can use multiple data items
  - E.g., testing different username and password combinations
  - Flexible in changing inputs
- Programming skills not needed when updating data
- Good for separating roles
  - Developers responsible for scripting
  - Testers responsible for test data

Data-Driven Testing [2]

- Disadvantages
  - Initial effort is required to create framework (parser, libraries, etc.)
    - Just like building a new tool
  - New tests need new driver scripts
  - Require programming skills to create new driver scripts
- Good solution for large scale automation

Keyword-Driven Testing [1]

- Separate test scripts from
  - test data
  - directives (keywords) on how to use data
- Keywords and data drive test execution

Keyword-Driven Testing [2]

- Advantages
  - All tests can be handled by one framework
  - Tests can be created from keywords
  - Non-programmers can create tests
  - Separate test data and scripts
- Disadvantages
  - Effort required to build framework upfront
  - Programming skills needed to build framework
- Good solution supported by many commercial tools

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Test Automation Tools

- Hundreds of test automation tools available in market
- Commercial tools
  - Usually powerful
  - Well supported
  - Rich feature set
  - Some are expensive, unluckily
- Open source
  - Free, of course
  - Many are good to use, others not that good q Support uncertain

Test Automation Tools

- Popular commercial tools q HP QTP/UFT
  - TestComplete
  - Ranorex
  - Rational Robot
  - Rational Functional Tester
  - eggPlant
- Popular open source and/or free tools
  - Selenium
  - Katalon Studio
  - Cucumber
  - Maveryx

Want to Work in Test Automation?

- Programming skills are a plus
  - Scripting languages: Python, Ruby, Java, C#, etc.
- Regular expressions
- SQL
- Learning to use test automation tools
  - Selenium
  - Katalon Studio

Conclusions

- Test automation is now a trend in software development
  - Customers require to apply test automation
  - Demand is high
- Test automation is essential for DevOps
- It does not replace manual testing